



Statistics of Financial Markets

Chapter 4: Probability



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Acronyms used in this material

- ▣ **AI:** Artificial Intelligence
- ▣ **AR:** AutoRegressive (model)
- ▣ **ARCH:** AutoRegressive Conditional Heteroskedasticity
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **CDF:** Cumulative Distribution Function
- ▣ **CLT:** Central Limit Theorem
- ▣ **CRR:** Cox–Ross–Rubinstein (binomial model)
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GBM:** Geometric Brownian Motion
- ▣ **IBM:** International Business Machines (company)
- ▣ **LCG:** Linear Congruential Generator
- ▣ **LLN:** Law of Large Numbers
- ▣ **PCG64:** Permuted Congruential Generator, 64-bit (NumPy default)
- ▣ **PMF:** Probability Mass Function
- ▣ **RANDU:** RANDU (IBM random number generator of the 1960s)
- ▣ **SFE:** Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)

Today's Question, Route and Learning Outcomes

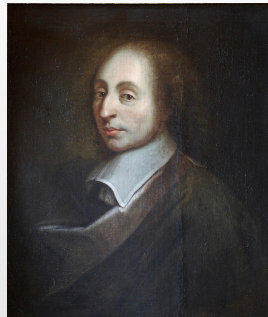
- **Question:** which probability tools do we need to describe returns and to simulate prices?
 - ▶ every later chapter (tails, volatility, VaR, efficiency) is built on them
- **Route and outcomes:** after this chapter you can
 - ▶ compute expectations, variances, covariances and correlations, and the volatility of a portfolio
 - ▶ explain why uncorrelated returns can still be dependent, and measure it
 - ▶ use conditional expectation and conditional variance, the building blocks of GARCH
 - ▶ generate random numbers with the inverse transform and estimate a probability by Monte Carlo, with its standard error
 - ▶ define and simulate the binomial model, a random walk, a martingale, white noise, AR(1), the Wiener process and GBM
- Real data, 2000–2026: S&P 500, DAX, BET and Bitcoin (from 2014)

Reading and Tools

- Textbook: [Franke, Härdle & Hafner \(2019\)](#), *Statistics of Financial Markets*, 5th ed., Ch. 3 (Sec. 3.1–3.5) and Ch. 4 (discrete-time stochastic processes)
 - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 3–4
- Monte Carlo methods in finance: [Glasserman \(2003\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_04](#)
 - ▶ ported from the SFE Quantlets of the textbook (SFErandu, SFErangen1, SFErangen2, SFEBinomp, SFEWienerProcess, SFEsimGBM)
 - ▶ each chart in these slides links to the Quantlet that draws it
- Lecture notebook: [open in Google Colab](#)
- Video course:  [Statistics of Financial Markets](#)

1654: the Problem of Points

- Two players stop a fair game before the end: how should the stake be split?
- Blaise Pascal and Pierre de Fermat solved it in letters (1654): split by the probability of winning, not by the score
- Example: A needs 1 more win, B needs 2; at most two more rounds
 - ▶ B wins only if B wins both rounds: probability $1/4$
 - ▶ A should get 75% of the stake: its **expected** gain
- The same idea prices a derivative today: a fair price is an expected payoff

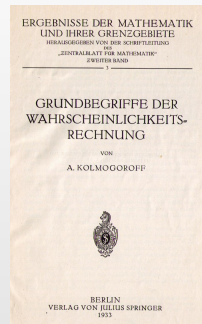


Blaise Pascal (1623–1662)

Portrait: unknown author, after François II Quesnel (c. 1690); public domain; Wikimedia Commons

Kolmogorov's Axioms (1933)

- A **probability space** $(\Omega, \mathcal{F}, P)$ Kolmogoroff (1933)
 - ▶ Ω : all possible outcomes (all possible paths of tomorrow's market)
 - ▶ $\mathcal{F}$: the events we can assign a probability to (a σ -algebra)
 - ▶ P : a function from events to $[0, 1]$
- Axioms, for any event $A \in \mathcal{F}$
 - ▶ $P(A) \geq 0$: no probability is negative
 - ▶ $P(\Omega) = 1$: some outcome certainly occurs
 - ▶ $P(\cup_i A_i) = \sum_i P(A_i)$ for disjoint $A_1, A_2, \dots$ (no two can occur together): the probability of "one of them" is the sum
- Everything else (expectation, independence, processes) is built on these three rules



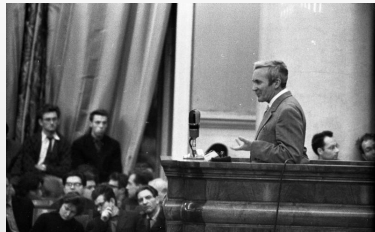
Title page of the *Grundbegriffe*, Springer 1933

Photo: KurtSchwitters (2012); CC BY-SA 3.0;

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Andrey Kolmogorov

- ▣ Andrey Nikolaevich Kolmogorov (1903–1987), Moscow State University
- ▣ Made probability a branch of measure theory: events are sets, probability is a measure
- ▣ Also: conditional expectation in its modern form, Markov processes, turbulence, complexity
- ▣ The Kolmogorov–Smirnov test ([Chapter 6](#)) carries his name



Kolmogorov lecturing in Moscow, 1963–1964

Photo: Vsevolod Tarasevich (1963–1964); CC BY 4.0; Wikimedia Commons

Events, Conditional Probability, Independence

- **Conditional probability:** $P(A | B) = P(A \cap B)/P(B)$, for $P(B) > 0$
 - ▶ A, B : two events; $A \cap B$: both occur
 - ▶ in words: among the cases in which B occurs, the share in which A also occurs
- **Bayes' rule:** $P(B | A) = P(A | B)P(B)/P(A)$
 - ▶ $P(B)$: the probability of B before we learn A ; $P(B | A)$: the updated probability once A is observed
 - ▶ it reverses the direction of conditioning: from “signal given crisis” to “crisis given signal”
- **Independent events:** $P(A \cap B) = P(A)P(B)$, i.e. $P(A | B) = P(A)$
 - ▶ learning that B occurred does not change the probability of A

Example: Down Days in a Row

- ▣ S&P 500, 2000–2026: $P(\text{down day}) = 46.3\%$
 - ▶ after a down day: $P(\text{down} \mid \text{down yesterday}) = 43.4\%$
 - ▶ two down days in a row: 20.1% in the data
 - ▶ under independence: $P(\text{down})^2 = 21.4\%$, close to the data
- ▣ BET: 50.0% after a down day vs 42.6% after an up day
 - ▶ the sign of the return persists from one day to the next
 - ▶ a consequence of thin trading: prices adjust to news over several days

Recap: From Games to Axioms

- ▣ Probability space $(\Omega, \mathcal{F}, P)$ and three axioms
- ▣ Fair value = expected value: the idea of Pascal and Fermat
- ▣ Conditional probability measures how information changes the odds

Random Variables

- A **random variable** $X : \Omega \rightarrow \mathbb{R}$ assigns a number to each outcome
 - ▶ tomorrow's log return, the number of up days in a week, the loss of a portfolio
- **Discrete**: finitely or countably many values x_i
 - ▶ **PMF** (probability mass function): $p(x_i) = P(X = x_i)$, with $\sum_i p(x_i) = 1$
 - ▶ $p(x_i)$: the probability of the value x_i ; the probabilities of all values add up to 1
- **Continuous**: $P(a < X \leq b) = \int_a^b f(x) dx$
 - ▶ the probability of an interval $(a, b]$ is the area under f between a and b
 - ▶ f is the **PDF** (probability density function): $f \geq 0$ and the total area $\int f = 1$
 - ▶ a single point has no area: $P(X = x) = 0$ for every x
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Sec. 3.1

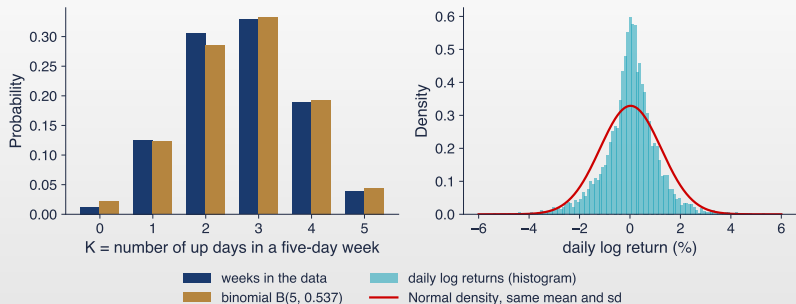
The Distribution Function and the Quantile Function

- ▣ **CDF** (cumulative distribution function): $F(x) = P(X \leq x)$
 - ▶ the probability that X does not exceed the threshold x
 - ▶ non-decreasing, right-continuous, from $F(-\infty) = 0$ to $F(\infty) = 1$
 - ▶ for a continuous X the density is its derivative: $f = F'$
- ▣ **Quantile** of order α : $q_\alpha = F^{-1}(\alpha) = \inf\{x : F(x) \geq \alpha\}$
 - ▶ $\alpha \in (0, 1)$: a probability; q_α is the smallest x with $P(X \leq x) \geq \alpha$
 - ▶ F^{-1} : the quantile function, the inverse of the CDF; $\inf$: the smallest value of the set
 - ▶ median = $q_{0.5}$; the 1% quantile is the return exceeded on 99% of the days

Value at Risk and the Empirical CDF

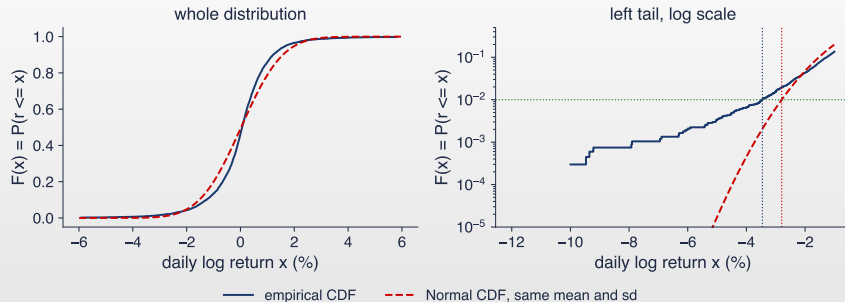
- **VaR** (Value at Risk) 1%: $\text{VaR}_{1\%} = -q_{0.01}$
 - ▶ $q_{0.01}$: the 1% quantile of the returns, a negative number
 - ▶ the minus sign turns it into a loss: the loss exceeded with probability 1%
 - ▶ estimation and backtesting in [Chapter 10](#)
- The **empirical CDF** of a sample: $\hat{F}(x) = \frac{1}{n} \#\{t : r_t \leq x\}$
 - ▶ $r_1, \dots, r_n$: the observed returns; n : the sample size
 - ▶ $\#\{\cdot\}$: the number of days in the set; $\hat{F}(x)$ is the share of days with a return at most x
 - ▶ the hat marks an estimate computed from data

A Discrete and a Continuous Random Variable



- Left: K = number of up days in a full week of the S&P 500 (1 144 weeks)
 - compared with $B(5, p)$, the binomial distribution: 5 days, each up with probability $p = 0.537$
 - mean 2.67 vs 2.69; variance 1.16 vs 1.24
- Right: daily log returns, a continuous random variable; the histogram estimates the PDF
- The binomial fits K well; the Normal density misses the peak and the tails of the returns ([Chapter 2](#))

Empirical CDF and the 1% Quantile of the S&P 500



- $n = 6714$ days; median 0.063%; $\hat{F}(0) = 46.3\%$ of the days are not up days
- VaR 1%: empirical 3.45%, Normal 2.80%; $P(r < -3\%)$: 1.55% vs 0.63%
- $P(r < -5\%)$: 0.34% in the data, 199 times the value under the Normal distribution ($1.7 \times 10^{-3}\%$)

Recap: Random Variables

- PMF for discrete, PDF for continuous variables; the CDF works for both
- Quantiles invert the CDF; VaR 1% is minus the 1% quantile
- The tails of the empirical CDF are much heavier than the Normal ones

Expectation

- **Expectation** (mean): $E[X] = \sum_i x_i p(x_i)$ or $E[X] = \int x f(x) dx$
 - ▶ $x_i, p(x_i)$: the values and their probabilities; f : the density
 - ▶ the centre of mass of the distribution; it exists only if $E|X| < \infty$ (not for Cauchy, [Chapter 3](#))
- Of a function: $E[g(X)] = \int g(x) f(x) dx$, without finding the distribution of $g(X)$
- **Linearity**: $E[aX + bY + c] = aE[X] + bE[Y] + c$, always (no independence needed)
 - ▶ a, b, c : constants; X, Y : any two random variables
 - ▶ the expected return of a portfolio is the weighted mean of the expected returns
- In general $E[g(X)] \neq g(E[X])$
 - ▶ Jensen: for a convex g , $E[g(X)] \geq g(E[X])$; for example $E[e^X] > e^{E[X]}$
 - ▶ the source of the volatility drag of [Chapter 1](#)

Variance and Moments

- **Variance:** $\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$
 - ▶ $\mu = E[X]$; the variance is the mean squared distance from the mean
 - ▶ standard deviation $\sigma = \sqrt{\text{Var}(X)}$, in the units of X ; for returns it is the volatility
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
 - ▶ adding a constant b does not change risk; multiplying by a multiplies the variance by a^2
- Moments: $E[X^k]$ for $k = 1, 2, \dots$
 - ▶ skewness $E[(X - \mu)^3]/\sigma^3$: the asymmetry of the distribution (0 for the Normal distribution)
 - ▶ kurtosis $E[(X - \mu)^4]/\sigma^4$: equals 3 for the Normal distribution
 - ▶ excess kurtosis = kurtosis $- 3$; positive for heavy tails ([Chapter 2](#))

Worked Example: Annual Mean and Volatility

- S&P 500, daily log returns r_t
 - ▶ daily mean 0.0247%, daily standard deviation $\sigma = 1.213\%$
 - ▶ $A = 252$: the number of trading days per year
- Assume i.i.d. daily returns (independent and identically distributed)
 - ▶ the annual log return is the sum of the A daily ones: $\sum_{t=1}^A r_t$
- Mean: $E[\sum r_t] = A \times 0.0247 = 6.2\%$ a year
 - ▶ linearity of expectation
- Variance: $\text{Var}(\sum r_t) = A\sigma^2$, because the covariances are zero
 - ▶ annual volatility $\sigma\sqrt{A} = 1.213\sqrt{252} = 19.3\%$
 - ▶ the square-root-of-time rule: volatility grows with $\sqrt{h}$ over h periods

How Rare Is a 4-Sigma Day? Chebyshev's Bound

- ▣ **Chebyshev's inequality:** for any X with finite variance, $P(|X - \mu| \geq k\sigma) \leq 1/k^2$
 - ▶ μ, σ : the mean and the standard deviation; $k > 1$: the distance from the mean, in standard deviations
 - ▶ it uses only the mean and the variance, so it holds for every distribution
- ▣ $k = 4$, S&P 500, 2000–2026
 - ▶ Chebyshev bound: at most 6.25% of the days
 - ▶ Normal distribution: 0.0063%
 - ▶ data: 46 days, 0.69%
- ▣ The data lie between the two
 - ▶ the Normal distribution is far too optimistic; the worst case of Chebyshev is too pessimistic

Recap: Expectation and Variance

- ▣ Expectation is linear; variance scales with the square
- ▣ For i.i.d. returns the mean grows with h and the volatility with $\sqrt{h}$
- ▣ Chebyshev bounds tail probabilities for any distribution

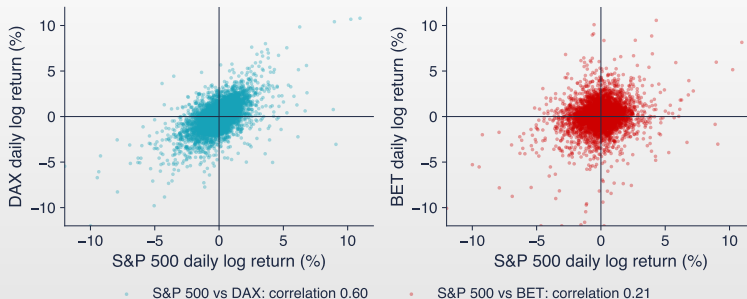
Joint Distribution and Marginals

- ▣ **Joint CDF:** $F(x, y) = P(X \leq x, Y \leq y)$
 - ▶ the probability that both variables are below their thresholds at the same time
 - ▶ for two returns: both markets fall below given levels on the same day
- ▣ **Marginal CDF:** $F_X(x) = F(x, \infty)$
 - ▶ the distribution of X alone, whatever the value of Y
- ▣ **Joint PDF** $f(x, y)$: $P((X, Y) \in B) = \iint_B f(x, y) dx dy$
 - ▶ B : a region of the plane; the probability is the volume under f above B
 - ▶ marginal density: $f_X(x) = \int f(x, y) dy$, integrating Y out

Covariance, Correlation and the Variance of a Sum

- **Covariance:** $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X\mu_Y$
 - ▶ μ_X, μ_Y : the means; positive when X and Y tend to be above their means together
- **Correlation:** $\rho = \text{Cov}(X, Y)/(\sigma_X\sigma_Y) \in [-1, 1]$
 - ▶ the covariance divided by the two standard deviations: a number without units
 - ▶ measures **linear** dependence only; $|\rho| = 1$ iff $Y = a + bX$
- **Variance of a sum:** $\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\text{Cov}(X, Y)$
 - ▶ a, b : constants (in a portfolio, the weights); the covariance term can raise or lower the risk
 - ▶ many assets: $\sigma_p^2 = w^\top \Sigma w$, with w the vector of weights and Σ the covariance matrix
 - ▶ σ_p^2 : the variance of the portfolio; $w^\top$: the transposed vector

Joint Distributions of Daily Returns



- Common trading days only: prices are joined first, then differenced ($n = 6\,605$ days for S&P 500 and DAX)
- S&P 500 and DAX: $\rho = 0.60$; S&P 500 and BET: $\rho = 0.21$
- The cloud is elliptic in the centre, but extreme days are joint: crises hit all markets together

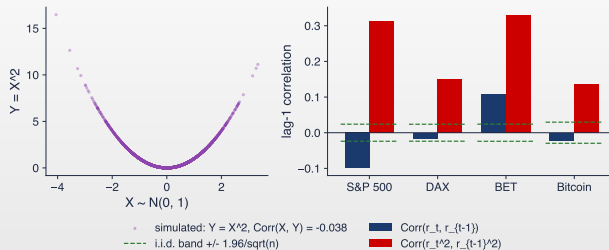
Worked Example: a 50/50 Portfolio of S&P 500 and DAX

- Daily standard deviations: $\sigma_1 = 1.22\%$ (S&P 500), $\sigma_2 = 1.43\%$ (DAX)
 - ▶ covariance 1.045; correlation 0.60
 - ▶ weights $a = b = 0.5$, so $a^2 = b^2 = 0.25$ and $2ab = 0.5$
- $\sigma_p^2 = 0.25\sigma_1^2 + 0.25\sigma_2^2 + 2 \times 0.25 \text{Cov} = 0.883 + 0.522 = 1.405$
 - ▶ $\sigma_p = 1.19\%$, below the average volatility 1.32%: diversification
- S&P 500 and BET, lower correlation: $\sigma_p = 1.04\%$ vs average 1.33%
 - ▶ the lower the correlation, the larger the reduction in risk
- Diversification works only through $\rho < 1$
 - ▶ in a crash correlations rise, exactly when diversification is needed

Independence and Uncorrelatedness

- X and Y are **independent** if $F(x, y) = F_X(x)F_Y(y)$ for all x, y
 - ▶ the joint CDF is the product of the marginal CDFs
 - ▶ then $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$ for **all** functions g, h (for example squares or absolute values)
- Independent $\Rightarrow$ uncorrelated ($g, h = \text{identity}$); the converse is false
- Example: $X \sim N(0, 1)$, $Y = X^2$
 - ▶ $\text{Cov}(X, X^2) = E[X^3] = 0$: uncorrelated
 - ▶ but Y is a function of X : completely dependent
- Exception: for jointly Normal variables, uncorrelated $\Leftrightarrow$ independent

Returns Are Uncorrelated, Squared Returns Are Not



Left: simulated $Y = X^2$, sample correlation -0.038 , yet Y is determined by X

Right: $\text{Corr}(r_t, r_{t-1})$ is small: S&P 500 -0.099 , DAX -0.017 , Bitcoin -0.022

► $\text{Corr}(r_t^2, r_{t-1}^2)$ is far outside the band: 0.31, 0.15, 0.14

► band $\pm 1.96/\sqrt{n}$: the values compatible with zero correlation at the 5% level

► the negative S&P 500 value comes mostly from crisis days

BET: $\text{Corr}(r_t, r_{t-1}) = 0.109$, an effect of thin trading

Recap: Joint Distributions

- Covariance enters the variance of every portfolio
- Independent implies uncorrelated, not the other way round
- Returns: nearly uncorrelated, but their squares are correlated, so they are not independent

Conditional Expectation

- Conditional density: $f(y | x) = f(x, y) / f_X(x)$
 - ▶ the distribution of Y once we know that $X = x$: the joint density rescaled by the marginal one
- **Conditional expectation:** $E[Y | X = x] = \int y f(y | x) dy$
 - ▶ the mean of Y computed with the conditional density
 - ▶ $E[Y | X]$ is a random variable, a function of X
- It is the best forecast of Y from X : it minimises $E[(Y - g(X))^2]$ over all functions g
 - ▶ $E[(Y - g(X))^2]$: the mean squared error of the forecast $g(X)$
 - ▶ for returns: $E[r_t | \mathcal{F}_{t-1}]$, the forecast given all information $\mathcal{F}_{t-1}$ up to yesterday
- **Tower property** (law of iterated expectations): $E[E[Y | X]] = E[Y]$
 - ▶ averaging the conditional forecasts over all values of X gives back the unconditional mean

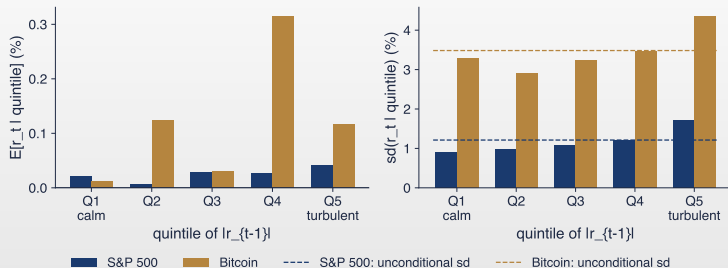
Conditional Variance and the Law of Total Variance

- **Conditional variance:** $\text{Var}(Y | X) = E[(Y - E[Y | X])^2 | X]$
 - ▶ the spread of Y around its conditional mean, once X is known
- **Law of total variance:** $\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])$
 - ▶ $E[\text{Var}(Y | X)]$: the average variance inside the groups defined by X (within)
 - ▶ $\text{Var}(E[Y | X])$: how much the group means differ from each other (between)
 - ▶ total risk = within-group variance + between-group variance

Worked Example: Two Regimes

- X = the market regime, $Y = r$ = the daily return (in %)
 - ▶ calm: probability 0.7, mean 0.05%, standard deviation 0.8%
 - ▶ turbulent: probability 0.3, mean -0.10% , standard deviation 2%
- Mean (tower property): $E[r] = 0.7 \times 0.05 + 0.3 \times (-0.10) = 0.005\%$
- The two parts of the variance
 - ▶ within: $E[\text{Var}(r \mid X)] = 0.7 \times 0.8^2 + 0.3 \times 2^2 = 1.648$
 - ▶ between: $\text{Var}(E[r \mid X]) = 0.7(0.05 - E[r])^2 + 0.3(-0.10 - E[r])^2 = 0.0047$
- $\text{Var}(r) = 1.653$, standard deviation 1.286%
 - ▶ almost all risk comes from the different variances of the regimes, not from their different means

Today's Return Given Yesterday's Move



- ▣ Quintiles of $|r_{t-1}|$: Q1 = the calmest 20% of yesterdays, Q5 = the most turbulent 20%
- ▣ Conditional mean: between 0.006% and 0.042% for the S&P 500, no usable pattern
- ▣ Conditional s.d.: S&P 500 from 0.91% (Q1) to 1.71% (Q5), ratio 1.88; Bitcoin ratio 1.33
- ▣ Law of total variance, S&P 500: $1.469 = 1.469 + 0.00013$; the between part is 0.01%

From Conditional Variance to GARCH

- Write the return as $r_t = \mu_t + \sigma_t z_t$, with z_t i.i.d., mean 0, variance 1
 - ▶ $\mu_t = E[r_t | \mathcal{F}_{t-1}]$: the conditional mean, close to constant
 - ▶ $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$: the conditional variance, which changes with the news
- ARCH (autoregressive conditional heteroskedasticity) Engle (1982): $\sigma_t^2 = \omega + \alpha r_{t-1}^2$
 - ▶ today's variance is a base level plus a share of yesterday's squared return
 - ▶ $\omega > 0$: the base level; $\alpha \geq 0$: how strongly a large move yesterday raises today's variance
- GARCH Bollerslev (1986): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ (Chapter 9)
 - ▶ $\beta \geq 0$: the persistence; a large β keeps a high variance high for many days
- Exactly the pattern in the chart: a large $|r_{t-1}|$ raises today's conditional s.d.
- The unconditional distribution is a mixture over σ_t
 - ▶ it has heavier tails than the Normal distribution, even if z_t is Normal

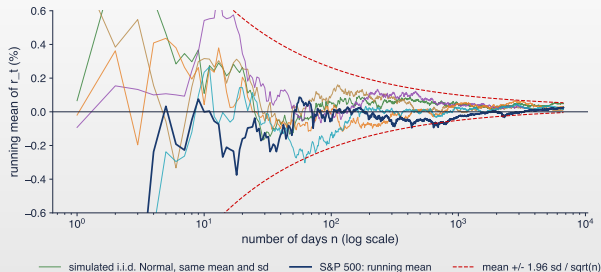
Recap: Conditioning

- ▣ $E[Y | X]$ is the best forecast; the tower property averages it back to $E[Y]$
- ▣ Returns: conditional mean almost flat, conditional variance strongly time-varying
- ▣ Modelling σ_t^2 is the job of ARCH and GARCH

Two Limit Theorems

- ▣ **LLN** (law of large numbers): for i.i.d. X_i with $E|X| < \infty$, $\bar{X}_n \rightarrow \mu$ as $n \rightarrow \infty$
 - ▶ $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$: the sample mean; $\mu = E[X]$: the true mean
 - ▶ the reason why averages, frequencies and Monte Carlo estimates work
- ▣ **CLT** (central limit theorem): with finite variance, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \rightarrow N(0, 1)$ ([Chapter 2](#))
 - ▶ the standardised sample mean has, for large n , approximately the standard Normal distribution
 - ▶ gives the speed: the error of $\bar{X}_n$ is about $\sigma/\sqrt{n}$
 - ▶ infinite variance: sums converge to α -stable laws instead ([Chapter 3](#))
- ▣ Both need some form of independence; dependent returns converge more slowly

The LLN for the Mean Return: Slow



- S&P 500, 26.6 years: mean 0.0247% a day, standard error $\sigma/\sqrt{n} = 0.0148\%$
 - ▶ $t = \text{mean} / \text{standard error} = 1.67$; $|t| > 1.96$ would mean a mean significantly different from 0 at 5%
- Annual mean 6.2%, 95% interval $[-1.1\%, 13.5\%]$
 - ▶ even 26 years do not pin down the expected return
 - ▶ for $t = 3$ at this mean and volatility we would need about 86 years of data

The Monte Carlo Idea

- Estimate $\theta = E[g(X)]$ by $\hat{\theta}_N = \frac{1}{N} \sum_{i=1}^N g(X_i)$ with simulated X_i
 - ▶ θ : the unknown quantity; g : a function of the simulated variable; N : the number of simulations
 - ▶ a probability is an expectation: $P(A) = E[1_A]$, with $1_A = 1$ if A occurs and 0 otherwise
 - ▶ the LLN makes $\hat{\theta}_N \rightarrow \theta$; the CLT gives the error $\text{sd}(g(X))/\sqrt{N}$
- Named after the Monaco casino by Ulam and von Neumann, Los Alamos, 1940s [Metropolis & Ulam \(1949\)](#)
- In finance [Glasserman \(2003\)](#)
 - ▶ prices of path-dependent options, portfolio VaR ([Chapter 10](#)), stress scenarios
 - ▶ any expectation without a closed form
- Two ingredients: uniform random numbers and a way to turn them into any distribution

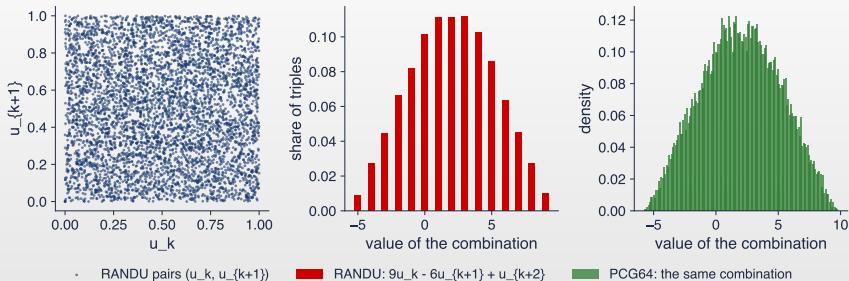
Pseudo-Random Numbers: Linear Congruential Generators

- **LCG** (linear congruential generator): $x_{k+1} = (ax_k + c) \bmod M$, $u_k = x_k/M \in [0, 1)$
 - ▶ each integer x_{k+1} is computed from the previous one x_k
 - ▶ a : the multiplier; c : the increment; M : the modulus; $\bmod M$: the remainder of the division by M
 - ▶ u_k : the uniform number in $[0, 1)$ obtained by dividing by M
- Deterministic: the same seed x_0 gives the same sequence
 - ▶ this makes results reproducible
 - ▶ the **period**: the number of values before the sequence repeats; at most M
- Real generators use huge M ; the period must be far longer than any simulation

Worked Example: an LCG with $M = 16$

- SFErangen1: $a = 5$, $c = 1$, $M = 16$, $x_0 = 7$
 - ▶ first step: $x_1 = (5 \times 7 + 1) \bmod 16 = 36 \bmod 16 = 4$
 - ▶ the sequence x : 7, 4, 5, 10, 3, 0, 1, 6, 15, 12, 13, 2, 11, 8, 9, 14, 7
- Period 16, the maximum possible, M
 - ▶ with $a = 4$ the sequence sticks after one step (period 1)
- Good uniform marginals are not enough
 - ▶ consecutive numbers must also look independent: the next chart

RANDU: Random Numbers Fall Mainly in the Planes (SFErandu)

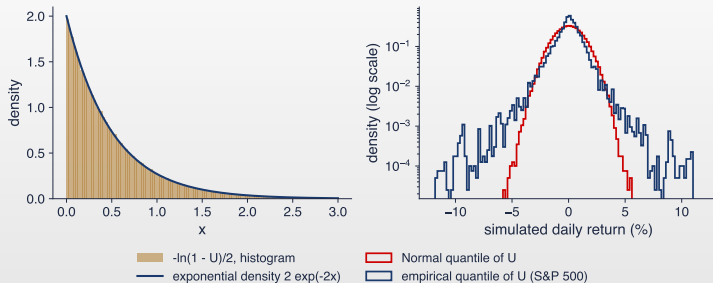


- RANDU, IBM, 1960s: $x_{k+1} = 65539 x_k \bmod 2^{31}$; pairs look uniform (left), correlation 0.003
- But $9u_k - 6u_{k+1} + u_{k+2}$ takes only 15 integer values, from -5 to 9 : all triples lie on 15 planes [Marsaglia \(1968\)](#)
- Modern generators: Mersenne Twister [Matsumoto & Nishimura \(1998\)](#), PCG64 (the NumPy default, right panel)

The Inverse Transform Method

- **Theorem:** if $U \sim U(0, 1)$ and F is a CDF, then $X = F^{-1}(U)$ has CDF F
 - ▶ $U(0, 1)$: the uniform distribution on $[0, 1]$, so $P(U \leq u) = u$
 - ▶ in words: a uniform number, read as a probability, is turned into the quantile of that order
 - ▶ proof: $P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$
- Exponential with rate $\lambda > 0$: $F(x) = 1 - e^{-\lambda x}$, so $X = -\ln(1 - U)/\lambda$
 - ▶ obtained by solving $u = F(x)$ for x ; the mean is $1/\lambda$
 - ▶ $\lambda = 2$, $u = 0.9$: $x = -\ln(0.1)/2 = 1.151$
- Normal: $X = \mu + \sigma \Phi^{-1}(U)$
 - ▶ Φ^{-1} : the quantile function of $N(0, 1)$; μ , σ : the desired mean and standard deviation
- Data: use the empirical quantile function $\hat{F}^{-1}$
 - ▶ this is resampling past days: historical simulation and the bootstrap

Inverse Transform: Exponential, Normal and Historical



- Left: 200 000 draws — $-\ln(1 - U)/2$ match the exponential density; sample mean 0.500 (theory $1/\lambda = 0.5$)
- Right: the same uniforms through the Normal and the empirical S&P 500 quantile functions
- Draws below -5% : 0.0035% Normal vs 0.34% historical (data: 0.34%); worst draw -5.7% vs -12.7%

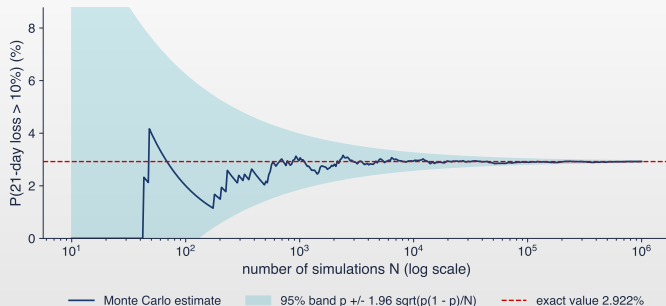
Worked Example: a Monthly Loss Probability by Monte Carlo (1/2)

- Model: S&P 500 daily log returns i.i.d. $N(0.0247, 1.213^2)$, in %
 - ▶ $N(m, s^2)$: the Normal distribution with mean m and variance s^2
- Event: the 21-day log return (one month) is below -10%
 - ▶ p : its probability, the quantity we want
- Exact: the sum of 21 i.i.d. Normal returns is $N(0.519, 5.56^2)$
 - ▶ mean 21×0.0247 , standard deviation $1.213\sqrt{21}$
 - ▶ standardise: $z = (-10 - 0.519)/5.56 = -1.892$
 - ▶ $p = \Phi(z) = 2.922\%$, with Φ the CDF of $N(0, 1)$

Worked Example: a Monthly Loss Probability by Monte Carlo (2/2)

- ▣ Monte Carlo: simulate N months; $\hat{p}_N$ = share of simulated months below -10%
 - ▶ standard error $\sqrt{p(1-p)/N}$: the typical distance between $\hat{p}_N$ and p
- ▣ Results
 - ▶ $N = 1\,000$: 3.000% (s.e. 0.533%)
 - ▶ $N = 100\,000$: 2.901% (s.e. 0.053%)
 - ▶ 10 times more precision needs 100 times more simulations
- ▣ For a relative error of 10% at the 95% level: $N \approx 1.96^2(1-p)/(0.1^2p) = 12\,762$
 - ▶ obtained from $1.96 \times \sqrt{p(1-p)/N} = 0.1p$: the half-width of the interval is 10% of p
 - ▶ a rare event needs many simulations

Monte Carlo Converges at Rate $1/\sqrt{N}$



- The estimate wanders for small N and settles inside the 95% band, which shrinks like $1/\sqrt{N}$
- $N = 10^6$: 2.926% vs exact 2.922%, s.e. 0.017%
- Monte Carlo removes the simulation error, not the model error: Seminar 4 (B3) repeats it with resampled days

Recap: Random Numbers and Monte Carlo

- ▣ Pseudo-random generators are deterministic; check them in several dimensions
- ▣ $F^{-1}(U)$ turns uniforms into any distribution, including the empirical one
- ▣ Monte Carlo error $\propto 1/\sqrt{N}$; always report it

The Binomial Process

- Each period the price is multiplied by u (up) with probability p , or by $d < u$ (down) with probability $1 - p$
 - ▶ u, d : the up and down factors (e.g. $u = 1.2$: +20%); the periods are independent
- After n steps with K up moves: $S_n = S_0 u^K d^{n-K}$
 - ▶ S_0 : the initial price; $K \sim B(n, p)$: the number of up moves is binomial
 - ▶ $\ln S_n = \ln S_0 + K \ln u + (n - K) \ln d$: a random walk in log prices (Section 9)
- $E[S_n] = S_0 (pu + (1 - p)d)^n$, because the steps are independent
 - ▶ $pu + (1 - p)d$: the expected growth factor of one step
 - ▶ the basis of the binomial option pricing model [Cox, Ross & Rubinstein \(1979\)](#); textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 7

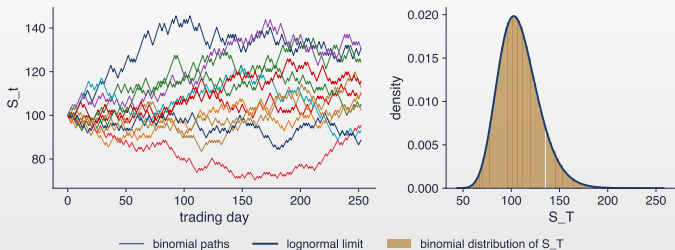
Worked Example: a Two-Step Tree

- $S_0 = 100$, $u = 1.2$, $d = 0.8$, $p = 0.6$
- Terminal values and probabilities
 - ▶ uu : 144 with $p^2 = 0.36$; ud or du : 96 with $2p(1-p) = 0.48$; dd : 64 with $(1-p)^2 = 0.16$
- $E[S_2] = 100 \times (0.6 \times 1.2 + 0.4 \times 0.8)^2 = 100 \times 1.04^2 = 108.16$
- $P(S_2 < 100) = 0.48 + 0.16 = 0.64$
 - ▶ one up and one down move lose money, because $ud = 0.96 < 1$
- With $p^* = (1-d)/(u-d) = 0.50$ the price is a fair game: $E[S_{t+1} | S_t] = S_t$ (Section 9)
 - ▶ p^* solves $p^*u + (1-p^*)d = 1$: the expected growth factor of a step equals 1

Calibrating the Tree to a Market (Cox–Ross–Rubinstein)

- **CRR** (Cox–Ross–Rubinstein) with step Δt : $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $p = (e^{\mu\Delta t} - d)/(u - d)$
 - ▶ Δt : the length of one step, in years; μ : the annual drift; σ : the annual volatility
 - ▶ $d = 1/u$: an up move followed by a down move returns to the start
 - ▶ chosen so that the mean and the variance of each step match a market with drift μ and volatility σ
- S&P 500, 2000–2026: $\sigma = 19.26\%$, $\mu = 8.08\%$ a year; daily steps, $\Delta t = 1/252$
 - ▶ $u = 1.0122$, $d = 0.9879$, $p = 0.5102$
- As $\Delta t \rightarrow 0$, $\ln S_T$ becomes Normal (CLT for K)
 - ▶ T : the horizon; the tree converges to GBM (Section 10)

Binomial Paths and the Lognormal Limit (SFEBinomp)



- Left: 12 paths over one year of daily steps, $S_0 = 100$
- Right: distribution of S_T after 252 steps ($T = 1$ year); $E[S_T] = 108.42 = 100e^\mu$
 - ▶ 5% quantile 78.5 vs 77.5 in the lognormal limit
- $P(S_T < 100)$: 39.7% binomial vs 37.3% lognormal
 - ▶ the exact value $S_T = 100$ has a positive probability, which the binomial counts on one side only

Recap: The Binomial Model

- Up or down each period: $S_n = S_0 u^K d^{n-K}$ with binomial K
- CRR matches μ and σ ; the limit is lognormal
- A special p^* turns the price into a fair game: the key to option pricing

Stochastic Processes and Information

- A **stochastic process** $\{X_t\}$: a family of random variables indexed by time
 - ▶ one realisation = one path; a price series is a single path of the process
- $\mathcal{F}_t$: the information available at time t (past prices, news)
 - ▶ it grows with t : nothing is forgotten (a **filtration**)
- **(Weakly) stationary**: $E[X_t]$ and $\text{Cov}(X_t, X_{t+h})$ do not depend on t
 - ▶ h : the lag between two dates; the mean and the dependence look the same at every date
 - ▶ returns: roughly yes; prices: no
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 4 and Sec. 11.3

White Noise and the Random Walk

- **White noise:** $E[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2$, $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$
 - ▶ ε_t : the shock at time t ; mean zero, constant variance σ^2 , no correlation across dates
 - ▶ i.i.d. noise is stronger: the ε_t are independent, not only uncorrelated
- **Random walk:** $S_t = S_{t-1} + \mu + \varepsilon_t = S_0 + \mu t + \sum_{s=1}^t \varepsilon_s$
 - ▶ each step adds the drift μ and a new shock; S_0 : the starting value
 - ▶ $E[S_t] = S_0 + \mu t$; $\text{Var}(S_t) = t\sigma^2$ grows with t : not stationary
 - ▶ simulated with $\sigma = 1$: variance 10.1, 49.8, 100.9 at $t = 10, 50, 100$
- Log prices as a random walk = returns as white noise
 - ▶ the benchmark of [Chapter 7](#) (efficient markets)

Martingales

- $\{X_t\}$ is a **martingale** if $E|X_t| < \infty$ and $E[X_{t+1} | \mathcal{F}_t] = X_t$
 - ▶ $E|X_t| < \infty$: the mean exists; $\mathcal{F}_t$: the information up to t
 - ▶ a fair game: the best forecast of tomorrow is today
- A **martingale difference**: $\varepsilon_t = X_t - X_{t-1}$ with $E[\varepsilon_t | \mathcal{F}_{t-1}] = 0$
 - ▶ the change cannot be predicted from the past
- Examples
 - ▶ a random walk without drift; $S_t^2 - t\sigma^2$ for the same walk; the binomial price under p^*
- Martingale $\neq$ i.i.d. increments: the conditional **variance** may change, as in GARCH
 - ▶ the efficient-market idea in its weak form **Fama (1970)**: excess returns (above the risk-free rate) are a martingale difference

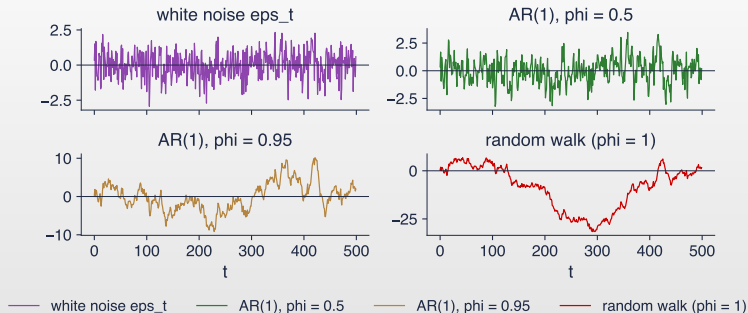
The AR(1) Process

- **AR(1)** (autoregressive of order 1): $X_t = c + \phi X_{t-1} + \varepsilon_t$, ε_t white noise
 - ▶ today's value is a constant, plus a share ϕ of yesterday's value, plus a new shock
 - ▶ c : the constant; ϕ : the persistence; σ^2 : the variance of ε_t
 - ▶ $\phi = 0$: white noise; $\phi = 1$, $c = \mu$: random walk
- Stationary iff $|\phi| < 1$; then
 - ▶ mean $c/(1 - \phi)$, variance $\sigma^2/(1 - \phi^2)$
 - ▶ autocorrelation at lag h : $\rho(h) = \text{Corr}(X_t, X_{t-h}) = \phi^h$, decaying geometrically
 - ▶ half-life of a shock: $\ln 0.5 / \ln \phi$ periods, the time until half of a shock is gone
- Uses
 - ▶ interest rates and spreads (mean reversion)
 - ▶ the squared returns behave like a persistent AR, which leads to GARCH

The AR(1) Process: Simulation

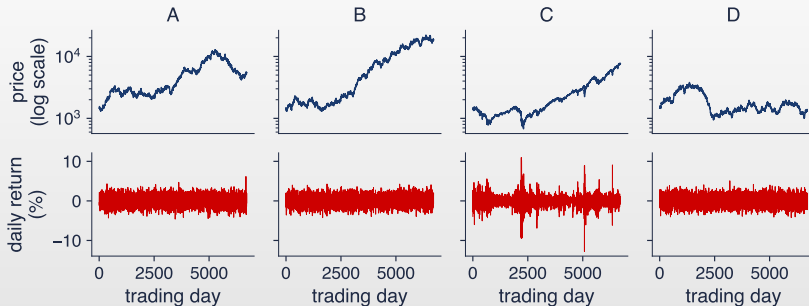
- Simulated with $\sigma = 1$, $\phi = 0.5$
 - ▶ variance 1.34 (theory $1/(1 - 0.5^2) = 1.33$)
 - ▶ $\rho(5) = 0.034$ (theory $0.5^5 = 0.031$); half-life 1.0 periods
- Simulated with $\sigma = 1$, $\phi = 0.95$
 - ▶ variance 10.20 (theory 10.26)
 - ▶ $\rho(5) = 0.772$; half-life 13.5 periods
- The closer ϕ is to 1, the larger the variance and the slower a shock fades

White Noise, AR(1) and Random Walk, Same Shocks



- The larger ϕ , the longer a shock lives: the AR(1) with $\phi = 0.95$ wanders, but returns to 0
- The random walk never returns on purpose: shocks are permanent and its variance grows with t

Question for the Room: Which One Is Real?



- One column is the S&P 500, 2000–2026; three are GBM paths (i.i.d. Normal returns) with the same mean and volatility
- **What do you think?** Which column is real, from the prices alone?
- **What do you think?** Which column is real, from the returns?

Answer: Column C

- **Answer:** column C is the S&P 500
- Prices: hard to tell; random walks with drift produce booms and long falls too
 - ▶ this is why [Bachelier \(1900\)](#) and [Osborne \(1959\)](#) modelled prices as random walks
- Returns: obvious; the real series has quiet years and violent clusters (2008, 2020)
 - ▶ excess kurtosis 10.7 vs at most 0.02 in the simulations; $\text{Corr}(r_t^2, r_{t-1}^2) = 0.31$ vs at most 0.006
 - ▶ largest absolute daily move 12.8% vs 6.2%
- Lesson: a model can fit the price path and still be wrong about risk

Recap: Discrete-Time Processes

- White noise: uncorrelated; i.i.d. noise: independent; a martingale difference: unpredictable mean
- Random walk: permanent shocks, variance $t\sigma^2$; AR(1) with $|\phi| < 1$: shocks fade at rate ϕ^h
- Prices look like random walks; returns are not i.i.d.

From Pollen to Prices: Brownian Motion

- 1827: the botanist Robert Brown sees pollen particles jiggling in water
- 1900: Louis Bachelier models the Paris bond market with the same motion, five years before physics [Bachelier \(1900\)](#)
- 1905: Albert Einstein explains it by molecular collisions [Einstein \(1905\)](#)
- 1923: Norbert Wiener builds the rigorous mathematical object [Wiener \(1923\)](#)
- 1959: [Osborne \(1959\)](#) applies it to log prices: geometric Brownian motion



Robert Brown (1773–1858)

Photo: [Mauil & Polyblank \(1855\)](#); public domain;
[Wikimedia Commons](#)

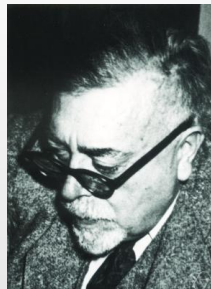


Louis Bachelier (1870–1946)

Photo: [unknown author](#); public domain; [Wikimedia Commons](#)

The Wiener Process

- $\{W(t), t \geq 0\}$ is a **Wiener process** (standard Brownian motion) if
 - ▶ $W(0) = 0$
 - ▶ increments over disjoint intervals are independent
 - ▶ $W(t) - W(s) \sim N(0, t - s)$ for $s < t$: the variance equals the elapsed time
 - ▶ the paths are continuous
- Limit of the scaled random walk $W_n(t) = (\varepsilon_1 + \dots + \varepsilon_{\lfloor nt \rfloor}) / \sqrt{n}$ (Donsker)
 - ▶ $\varepsilon_i = \pm 1$ with probability $1/2$ each; n steps per unit of time
 - ▶ $\lfloor nt \rfloor$: the number of steps up to time t (integer part); dividing by $\sqrt{n}$ keeps the variance equal to t
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 5

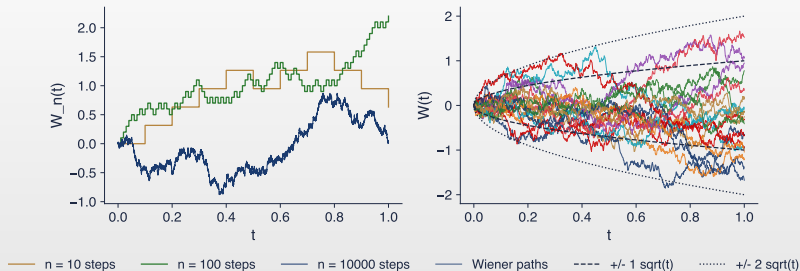


Norbert Wiener (1894–1964)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE;

[Wikimedia Commons](#)

From Random Walks to the Wiener Process (SFEWienerProcess)



Left: scaled random walks with 10, 100 and 10 000 steps; the steps vanish, the randomness stays

Right: 20 Wiener paths; $\text{Var}(W(t)) = t$, so they spread like $\pm\sqrt{t}$

► simulated $\text{Var}(W(1)) = 1.006$; 68.2% of $W(1)$ inside ± 1 (theory 68.3%)

Strange Properties of Wiener Paths

- Continuous everywhere, differentiable nowhere
 - ▶ the slope $(W(t+h) - W(t))/h \sim N(0, 1/h)$ has a variance $1/h$ that grows without bound as $h \rightarrow 0$
- **Quadratic variation:** $\sum_k (W(t_{k+1}) - W(t_k))^2 \rightarrow t$ on a finer and finer grid
 - ▶ $0 = t_0 < t_1 < \dots < t$: the grid; the sum of squared increments tends to the elapsed time t
 - ▶ 1000 steps on $[0, 1]$: mean 0.999, s.d. 0.045 across 10 000 paths
 - ▶ the reason why $(dW)^2 = dt$ in Itô calculus, and why realised variance measures volatility ([Chapter 8](#))
- $\text{Cov}(W(s), W(t)) = \min(s, t)$: simulated $\text{Cov}(W(0.5), W(1)) = 0.496$
- W is a martingale; so is $W(t)^2 - t$

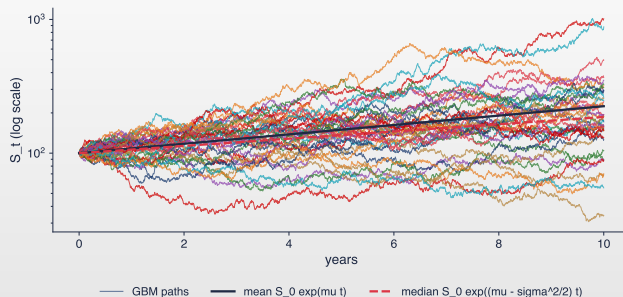
Geometric Brownian Motion

- **GBM** (geometric Brownian motion): $dS_t = \mu S_t dt + \sigma S_t dW_t$
 - ▶ dS_t : the change of the price over a very short interval dt
 - ▶ dW_t : the increment of the Wiener process over dt , distributed $N(0, dt)$
 - ▶ the relative change dS/S has drift μ and volatility σ , both per year
- Solution: $S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t)$
 - ▶ $\ln(S_t/S_0) \sim N((\mu - \sigma^2/2)t, \sigma^2 t)$
 - ▶ lognormal prices, i.i.d. Normal log returns
- The model behind [Black & Scholes \(1973\)](#) ([Chapter 6](#) of the textbook)

GBM: Mean, Median and Simulation

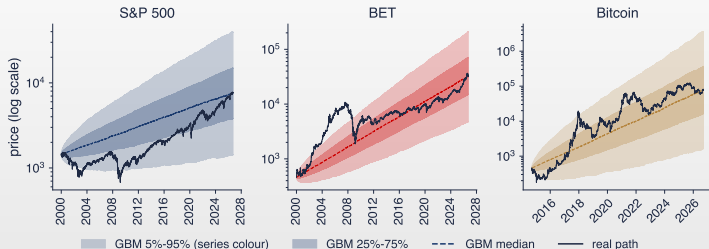
- Mean $E[S_t] = S_0 e^{\mu t}$; median $S_0 e^{(\mu - \sigma^2/2)t}$
 - ▶ the median is lower: half of the paths end below $S_0 e^{(\mu - \sigma^2/2)t}$
 - ▶ the gap, $\sigma^2/2$ a year in log terms, is the volatility drag
- Exact simulation on a grid: $S_{t+\Delta t} = S_t \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} Z)$
 - ▶ Δt : the grid step, in years; $Z \sim N(0, 1)$, a new independent draw at each step
 - ▶ exact because each log increment is Normal with the right mean and variance

GBM Calibrated to the S&P 500 (SFEsimGBM)



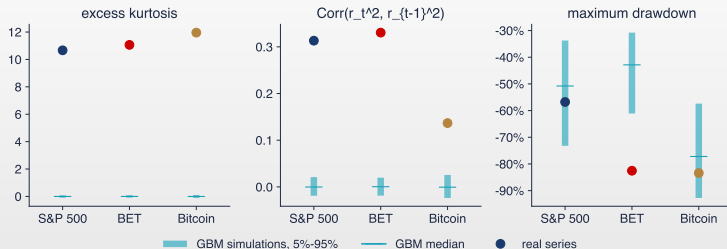
- ▣ Calibration, 2000–2026: $\sigma = 19.26\%$, log drift $\mu - \sigma^2/2 = 6.23\%$, so $\mu = 8.08\%$ a year
- ▣ After $T = 10$ years: mean 224.4, median 186.4
 - ▶ 62.0% of the paths end below the mean; theory: $\Phi(\sigma\sqrt{T}/2) = 62.0\%$, with Φ the $N(0, 1)$ CDF
- ▣ Probability of a loss after 10 years: 15.3%

Real Paths in the GBM Fan



- ▣ Each fan: 4000 GBM paths with the drift and volatility of the series itself, started on its first day
- ▣ S&P 500: the path reached the 0.5% quantile on 23 July 2002, then the median at the end
 - ▶ outside the 90% band 7% of the time
- ▣ BET: outside the band 24% of the time
 - ▶ the 2000–2007 boom and the 2008 crash are not GBM-like
- ▣ Bitcoin: $\sigma = 66.6\%$ a year makes the fan enormous
 - ▶ annualised with $A = 365$: crypto trades every calendar day ($A = 252$ for the indices)

What GBM Misses

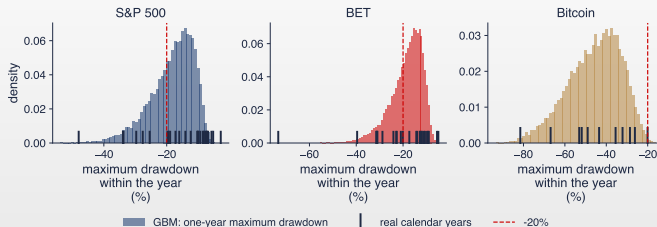


- Excess kurtosis: S&P 500 10.7, BET 11.1, Bitcoin 12.0; 90% of 300 GBM simulations lie within $[-0.10, 0.09]$
- Volatility clustering: $\text{Corr}(r_t^2, r_{t-1}^2) = 0.31, 0.33, 0.14$ vs at most 0.03 under GBM
- Maximum drawdown $\min_t (S_t / \max_{s \leq t} S_s - 1)$: the largest fall from a previous peak
 - S&P 500 -57% is ordinary for GBM; BET -83% is beyond all but 5% of the paths (5% quantile -61%)
- 4-sigma days: 46 (S&P 500) vs at most 1 in 95% of the GBM paths; crypto has the same problem [Pele et al. \(2023\)](#)

Recap: Wiener Process and GBM

- Wiener process: independent Normal increments with variance equal to the elapsed time
- GBM: lognormal prices, median below mean, the benchmark of option pricing
- Real returns have heavy tails and volatility clusters that GBM cannot produce

An Open Question: Does GBM Get Drawdown Risk Right?



- Drawdown (What GBM Misses), here within a calendar year
 - ▶ beyond 20%: S&P 500 in 6 of 26 years (23%) vs 34% under GBM; BET 46% vs 34%
- Beyond 40%: GBM expects 0.14 such years for the S&P 500 and 0.18 for the BET
 - ▶ both had 1 (2008: -48% and -73%)
- Open question
 - ▶ does GBM overstate moderate drawdowns and understate extreme ones because calm and crisis years cluster?

How AI Could Help

- **Literature:** find studies of drawdown distributions and of their link to volatility clustering, and summarise them
- **Code:** draft simulations of yearly drawdowns under GBM, the i.i.d. bootstrap and a GARCH model
- **Design:** propose a test that compares the observed number of bad years with the model probability
- **Example prompt**
 - ▶ Write Python code that simulates 10,000 one-year paths of daily S&P 500 returns under GBM and under an i.i.d. bootstrap of 2000-2026 returns, and returns the probability of a drawdown beyond 20% and 40% within the year.

What to Check

- ▣ Definitions: drawdown from the running peak within the year, not from the first day of the year
- ▣ Calibration: the same μ and σ , the same number of trading days a year as the data
- ▣ Inference: 26 years is a small sample; compute a binomial p-value, not only shares
- ▣ Mechanism: the i.i.d. bootstrap keeps heavy tails but removes clustering; the difference isolates the effect of clustering
- ▣ References: every cited paper must exist; check the DOI

Project Seed

- **Question:** how wrong is GBM about the probability of a bad year, and why?
 - ▶ data: S&P 500, DAX, BET and Bitcoin from the course data
- Steps
 - ▶ compute within-year maximum drawdowns for every calendar year
 - ▶ simulate the same statistic under GBM, the i.i.d. bootstrap and a GARCH(1,1) model ([Chapter 9](#))
 - ▶ compare the shares beyond 20% and 40% with binomial tests
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected [Wang et al. \(2023\)](#)

Key Formulas

Quantity	Formula
Variance of a sum	$\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\text{Cov}(X, Y)$
Correlation	$\rho = \text{Cov}(X, Y)/(\sigma_X\sigma_Y)$
Tower property	$E[E[Y X]] = E[Y]$
Total variance	$\text{Var}(Y) = E[\text{Var}(Y X)] + \text{Var}(E[Y X])$
Inverse transform	$X = F^{-1}(U), U \sim U(0, 1)$
Monte Carlo error	$\sqrt{p(1-p)/N}$
Random walk	$S_t = S_{t-1} + \mu + \varepsilon_t, \text{Var}(S_t) = t\sigma^2$
AR(1)	$\text{Var}(X_t) = \sigma^2/(1 - \phi^2), \rho(h) = \phi^h$
GBM	$S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t), E[S_t] = S_0 e^{\mu t}$

Check Yourself

- **Question:** $\sigma_1 = 2\%$, $\sigma_2 = 3\%$, $\rho = 0.5$; what is the volatility of the 50/50 portfolio?
 - ▶ **Answer:** $\sqrt{0.25 \times 4 + 0.25 \times 9 + 2 \times 0.25 \times 0.5 \times 2 \times 3} = \sqrt{4.75} = 2.18\%$
- **Question:** GBM with $\mu = 8\%$, $\sigma = 20\%$, $S_0 = 100$; what are the mean and the median of S_1 ?
 - ▶ **Answer:** mean $100e^{0.08} = 108.33$, median $100e^{0.06} = 106.18$
- **Question:** AR(1) with $\phi = 0.8$ and $\sigma = 1$; what is its variance?
 - ▶ **Answer:** $1/(1 - 0.64) = 2.78$
- Next: [Chapter 5](#), heavy tails and extreme value theory

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